Bank Customer Churn Prediction

End-to-End Machine Learning Project Report

Dataset: Churn_Modelling.csv | 10,000 customer records | 14 features

1. Executive Summary

This report presents an end-to-end machine learning analysis of bank customer churn using a dataset of 10,000 customers across France, Germany, and Spain. The goal is to identify which customers are likely to close their accounts ("Exited") so the bank can intervene with retention strategies before they leave.

Three classification models were trained and compared: Logistic Regression, Decision Tree, and Random Forest. The Random Forest model delivered the strongest overall performance, reaching 86.4% accuracy and an F1 score of 0.579 on the churn class, and is recommended as the production model.

Best Model	Accuracy	Churn Rate
Random Forest	86.4%	20.4%

2. Dataset Overview

The dataset contains 10,000 rows and 14 columns, with no missing values. The target variable, Exited, indicates whether a customer closed their account (1) or remained (0).

- Total customers: 10,000
- Churned (Exited = 1): 2,037 customers (20.4%)
- Retained (Exited = 0): 7,963 customers (79.6%)
- Identifier columns dropped before modeling: RowNumber, CustomerId, Surname
- Categorical features encoded: Geography, Gender (Label Encoding)

The class imbalance (~80/20 split) means accuracy alone is not a sufficient metric — F1 score, precision, and recall on the churn class are more informative, and a stratified train/test split (80/20) was used to preserve this ratio.

3. Exploratory Data Analysis

3.1 Churn Distribution

Roughly 1 in 5 customers in the dataset has churned, confirming a moderate class imbalance that motivates the use of precision/recall/F1 alongside accuracy.

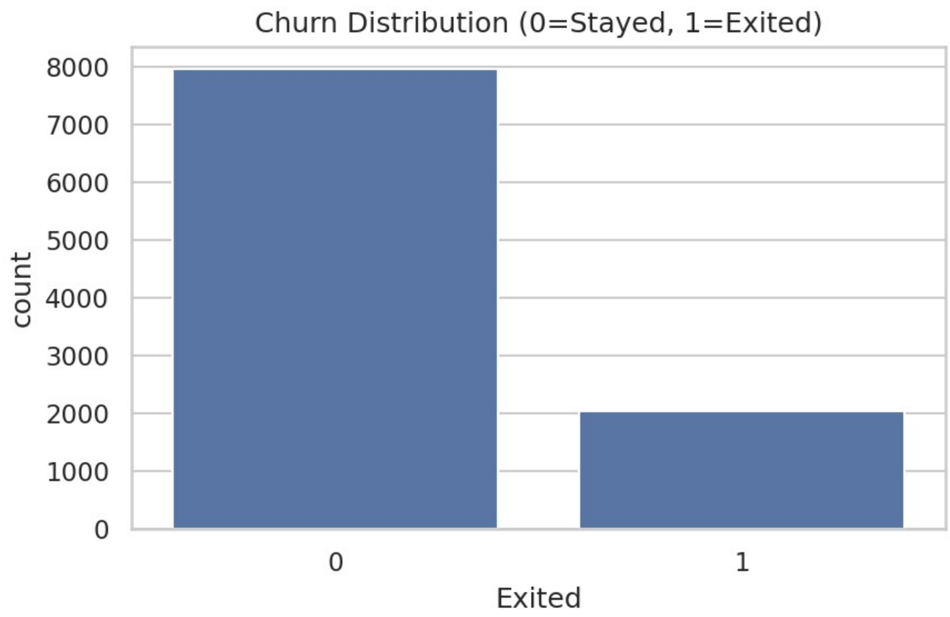

Figure 1. Overall churn distribution (0 = Stayed, 1 = Exited)

3.2 Age vs Churn

Churned customers skew notably older than retained customers, with a higher median age and a tighter, upper-shifted interquartile range — suggesting age is a strong behavioral signal.

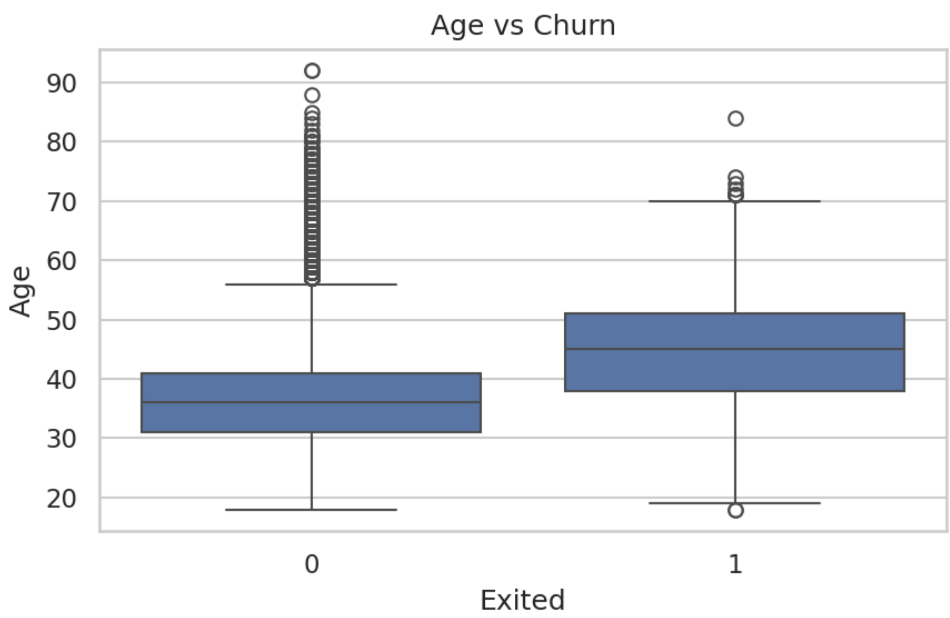

Figure 2. Age distribution by churn status

3.3 Churn by Geography

Germany shows a disproportionately high churn rate relative to its customer base size compared to France and Spain, despite France having the most customers overall.

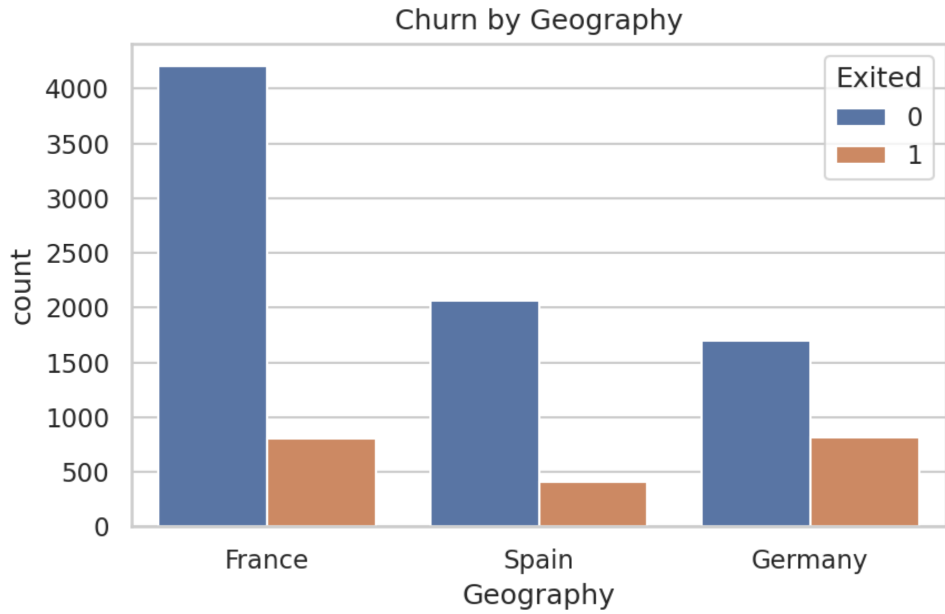

Figure 3. Churn counts by country

3.4 Churn by Gender

Female customers churn at a somewhat higher rate than male customers in this dataset.

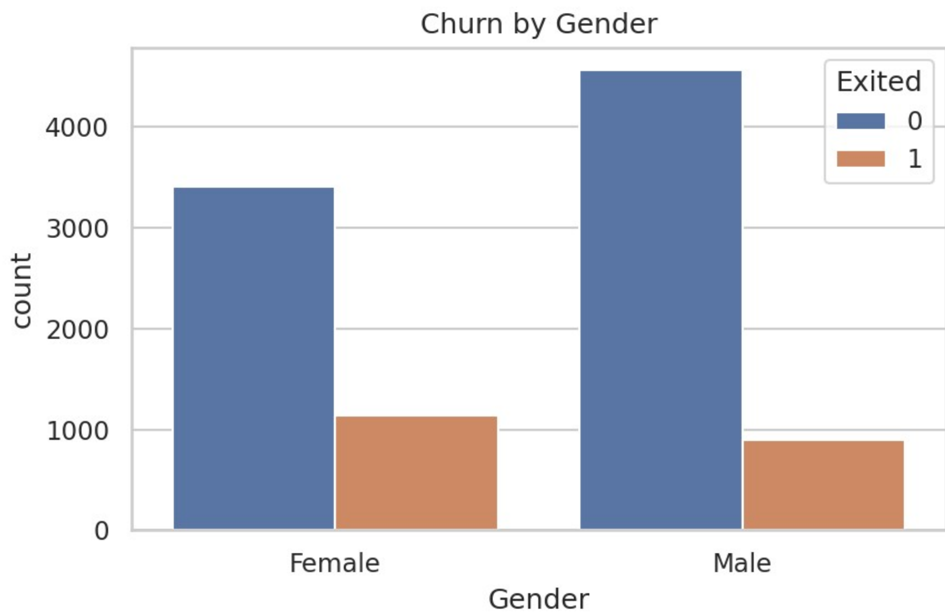

Figure 4. Churn counts by gender

3.5 Credit Score vs Churn

Credit score distributions are broadly similar between churned and retained customers, with only a slightly lower median for churned customers — indicating credit score alone is a weak standalone churn signal.

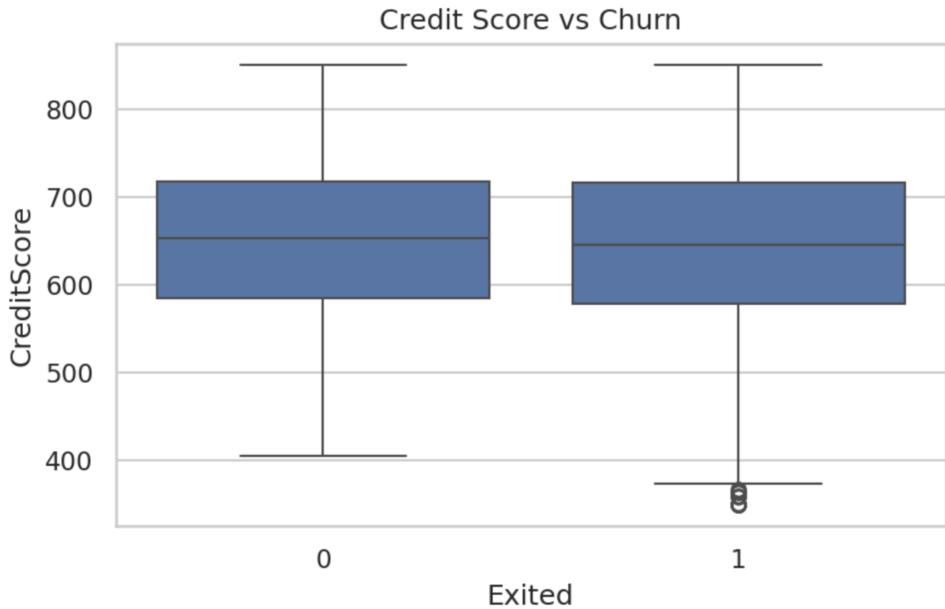

Figure 5. Credit score distribution by churn status

3.6 Number of Products

Customers holding exactly one product churn at a noticeably higher rate, while customers with 3 or 4 products show a sharply elevated churn rate despite being a small group — a strong candidate risk segment worth investigating further (potentially product dissatisfaction or over-selling).

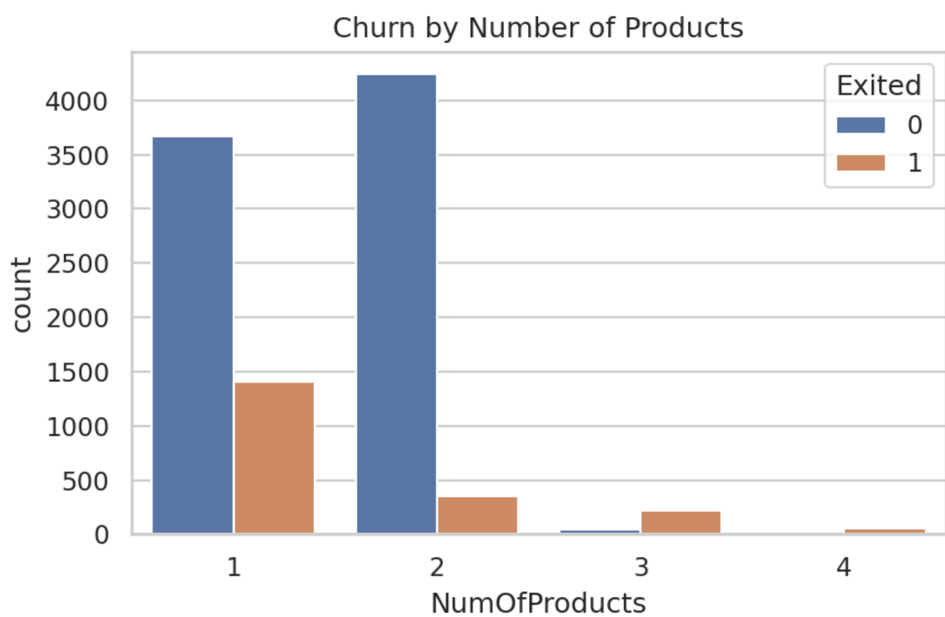

Figure 6. Churn by number of bank products held

3.7 Active Membership

Inactive members churn at roughly double the rate of active members, making activity status one of the clearer behavioral churn indicators.

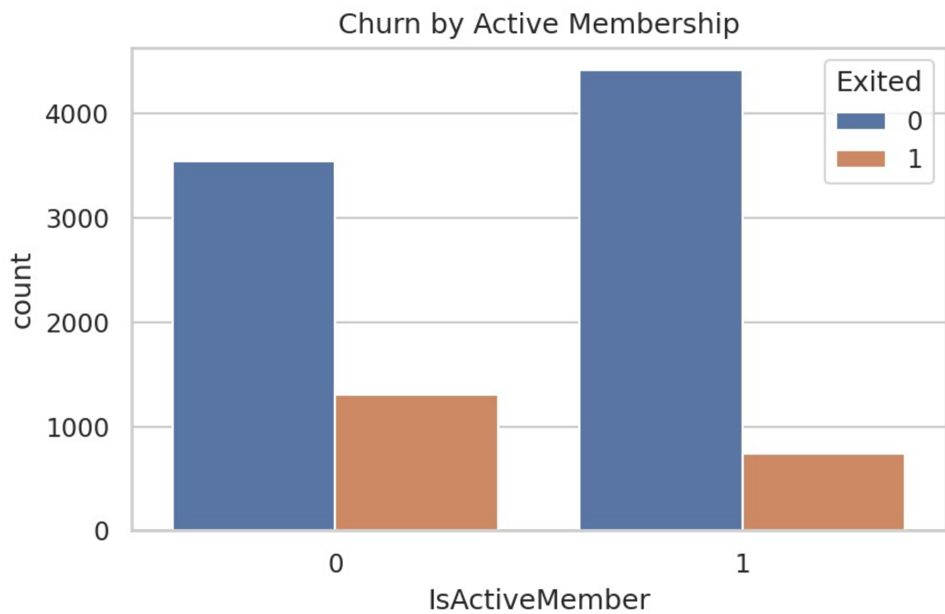

Figure 7. Churn by active membership status

3.8 Correlation Heatmap

No individual numeric feature shows a strong linear correlation with Exited; Age has the highest (mild) positive correlation, while IsActiveMember shows a mild negative correlation. This weak linear structure explains why non-linear models (Decision Tree, Random Forest) outperform Logistic Regression.

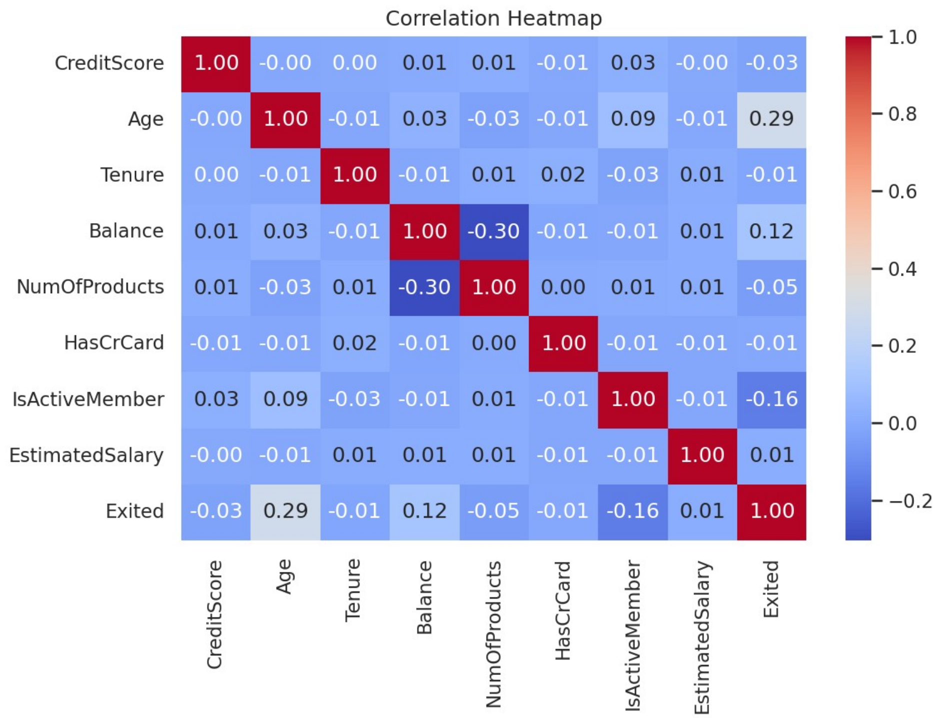

Figure 8. Correlation matrix of numeric features

4. Data Preprocessing

- Dropped non-predictive identifier columns: RowNumber, CustomerId, Surname
- Label-encoded Geography and Gender into numeric codes
- Split data 80/20 into training and test sets, stratified on Exited to preserve class balance
- Standardized features (zero mean, unit variance) for Logistic Regression only; tree-based models used raw feature values since they are scale-invariant

5. Model Training & Comparison

Three classifiers were trained and evaluated on the held-out 20% test set (2,000 customers):

- Logistic Regression (max_iter=1000, on scaled features)
- Decision Tree (default depth, random_state=42)
- Random Forest (100 trees, random_state=42)

Model	Accuracy	F1 Score	Precision	Recall	RMSE	ROC AUC
Random Forest	0.864	0.579	0.782	0.459	0.369	0.846
Decision Tree	0.776	0.465	0.453	0.477	0.473	0.665
Logistic Regression	0.805	0.229	0.586	0.143	0.442	0.771

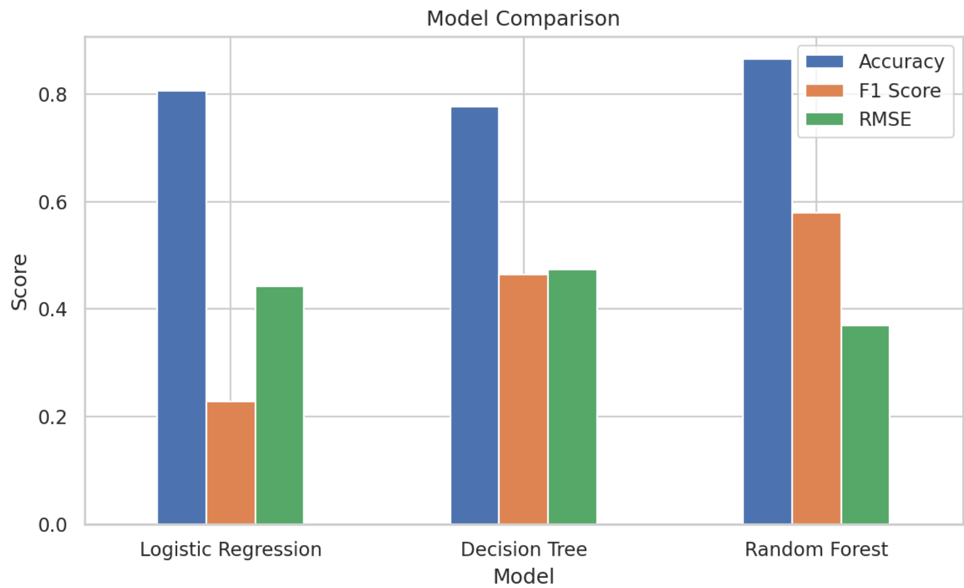

Figure 9. Accuracy, F1 score, and RMSE across models

5.1 Random Forest — Best Performer

Random Forest achieved the highest accuracy (86.4%), F1 score (0.579), precision (78.2%), and ROC AUC (0.846). It correctly identifies churners far more reliably than the other two models while keeping false positives relatively low.

5.2 Decision Tree

The single Decision Tree achieved the highest recall (47.7%) among the three models but at a cost of much lower precision (45.3%) and overall accuracy (77.6%), reflecting overfitting to training-set noise typical of unpruned trees.

5.3 Logistic Regression

Logistic Regression had the weakest churn-detection performance (F1 = 0.229, recall = 14.3%) despite reasonable overall accuracy (80.5%). Its linear decision boundary struggles to capture the non-linear interactions between features (e.g., product count and activity status) that drive churn.

6. ROC Curve Comparison

The ROC curve confirms Random Forest's superiority, with the largest area under the curve (AUC = 0.846), followed by Logistic Regression (0.771) and Decision Tree (0.665). Random Forest's ensemble averaging reduces the overfitting seen in the single Decision Tree while still capturing non-linear patterns Logistic Regression misses.

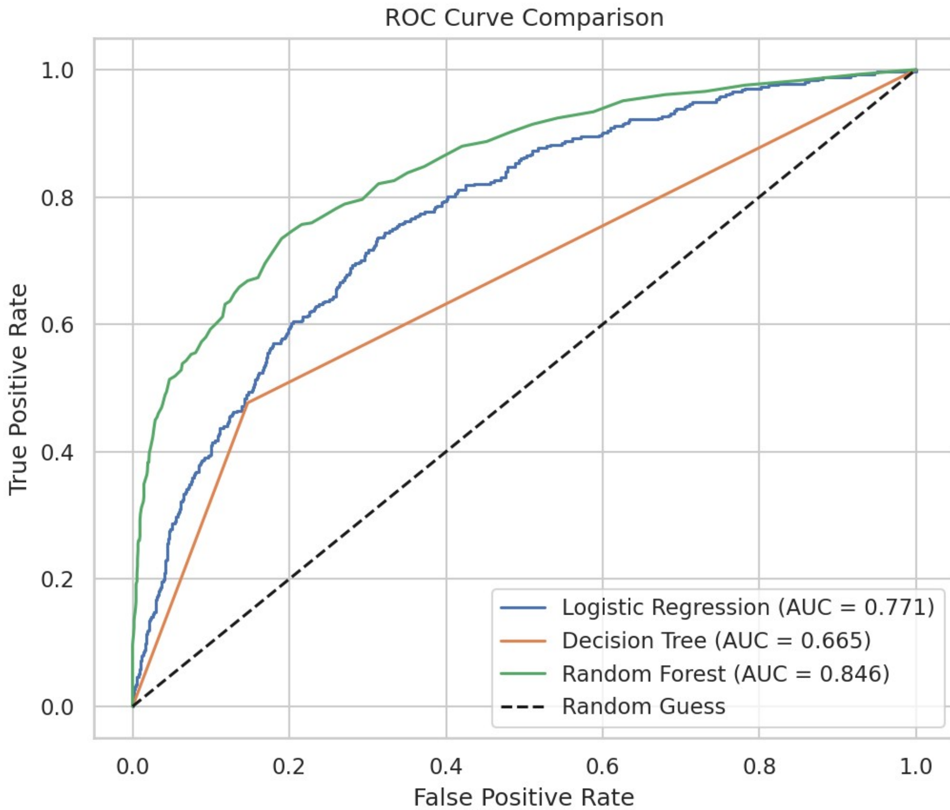

Figure 10. ROC curves for all three models

7. Feature Importance (Random Forest)

Age is the single strongest predictor of churn, followed by a cluster of comparably important features: EstimatedSalary, Balance, and CreditScore. Behavioral/relationship features (NumOfProducts, Tenure, IsActiveMember) also contribute meaningfully, while demographic flags like HasCrCard and Gender contribute the least.

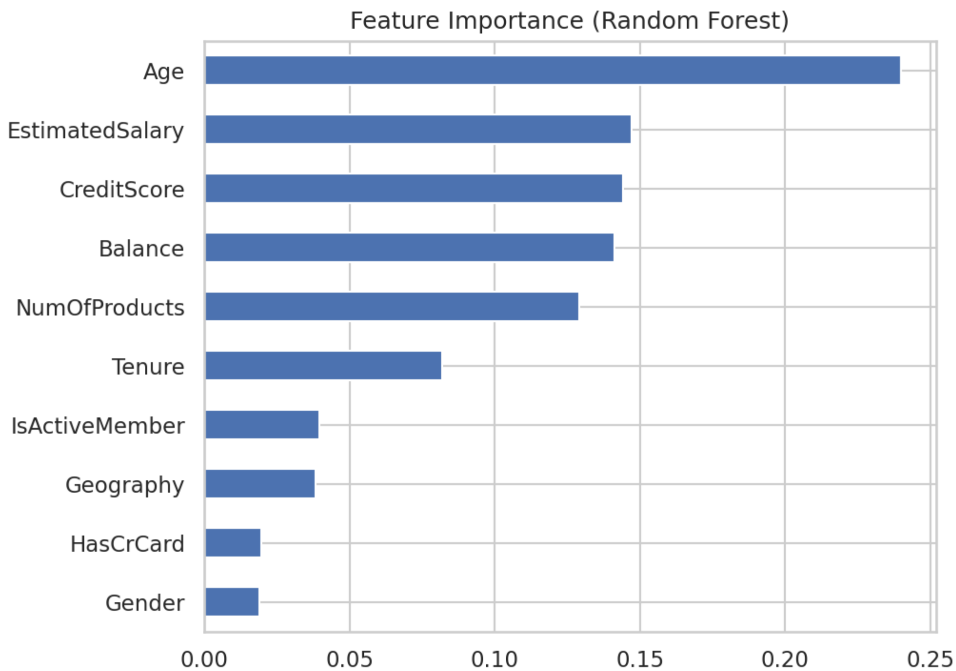

Figure 11. Random Forest feature importances

Feature	Importance
Age	0.238
NumOfProducts	0.132
Balance	0.145
EstimatedSalary	0.145
CreditScore	0.143
IsActiveMember	0.043
Tenure	0.081
Geography	0.030
Gender	0.017
HasCrCard	0.017

8. Confusion Matrices

Confusion matrices for each model on the 2,000-row test set (1,593 stayed, 407 exited):

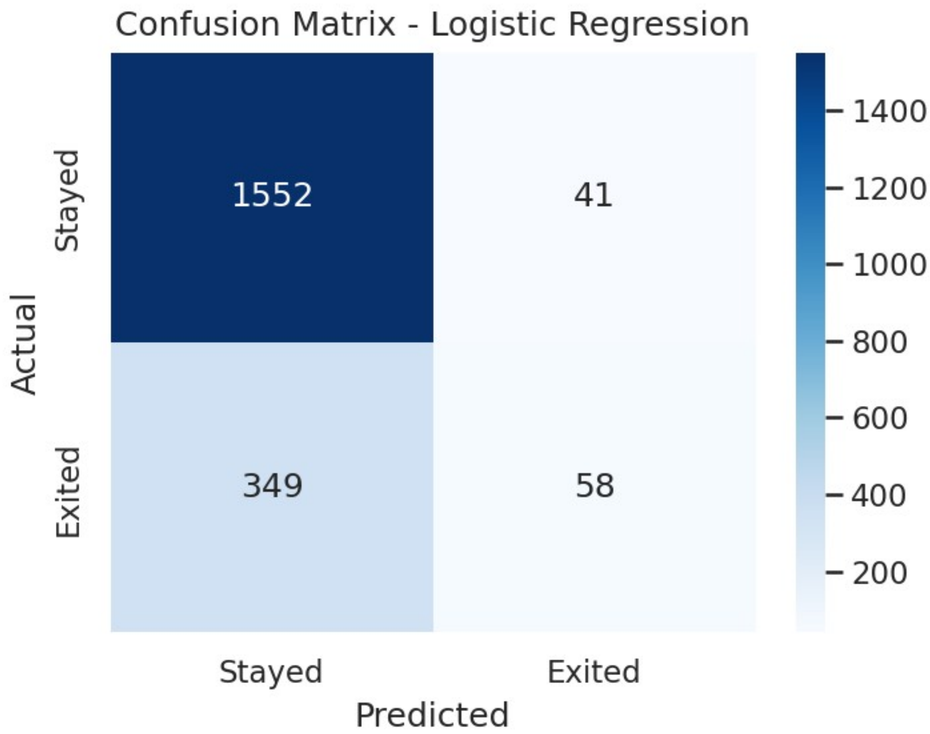

Figure 12. Logistic Regression confusion matrix

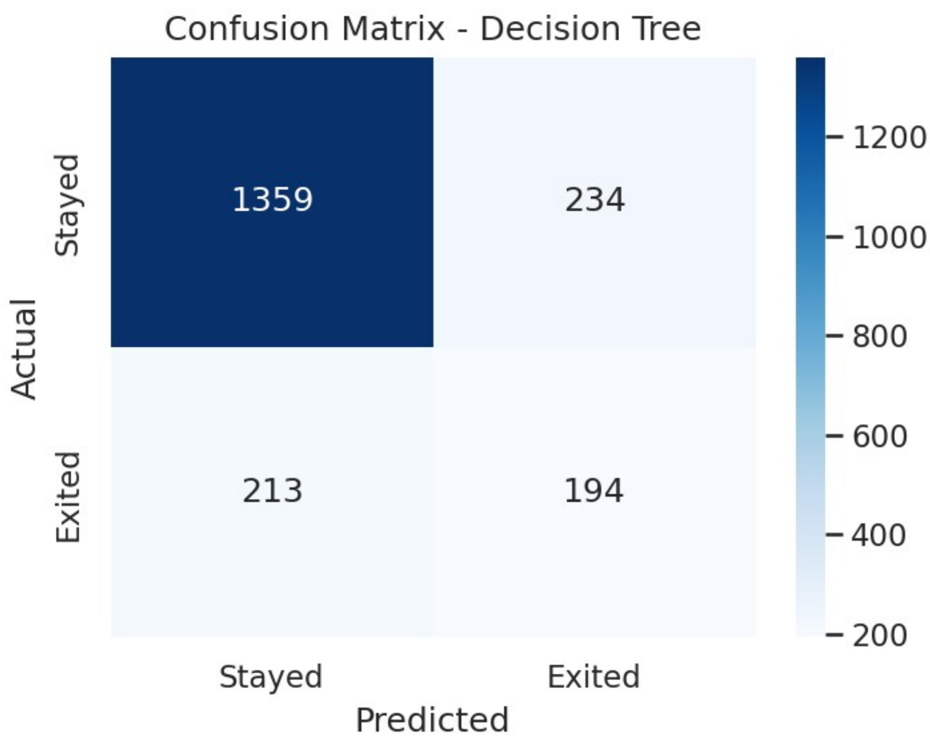

Figure 13. Decision Tree confusion matrix

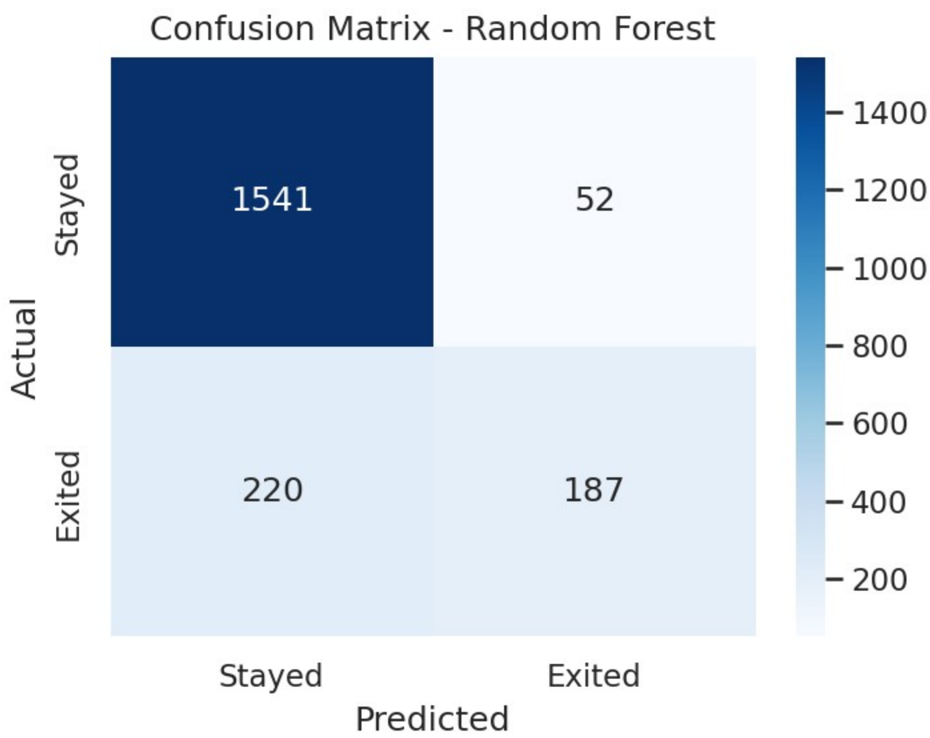

Figure 14. Random Forest confusion matrix

Random Forest correctly identifies 187 of 407 actual churners while keeping false positives low (52), giving it the best balance of precision and recall of the three models. Decision Tree catches slightly more churners (194) but at the cost of far more false alarms (234), which would translate into a much less efficient retention campaign.

9. Key Insights

- Older customers are meaningfully more likely to churn — retention campaigns should prioritize customers in higher age brackets.
- Inactive members churn at roughly 2x the rate of active members — re-engagement campaigns (app usage prompts, incentives) could reduce churn.
- Customers with 3–4 products churn disproportionately, despite being a small segment — worth a targeted investigation into whether these customers were over-sold or are dissatisfied.
- Germany shows a higher relative churn rate than France or Spain — a market-specific retention strategy may be warranted.
- Random Forest is the most reliable model in this comparison for identifying at-risk customers and should be the basis for any production churn-scoring system.

10. Recommendations

- Deploy the Random Forest model as the churn-scoring engine, generating a monthly risk score per customer.
- Prioritize retention outreach (relationship manager calls, tailored offers) for customers aged 45+ who are inactive or hold 3+ products.
- Investigate the root cause of elevated churn among multi-product holders — likely a service, pricing, or onboarding issue.

- Consider gathering additional behavioral features (transaction frequency, complaint history, app engagement) to further boost recall on the churn class, since the current best F1 (0.579) still leaves room for improvement.
- Re-evaluate class-imbalance handling (e.g., class weighting or SMOTE) in a future iteration to push recall higher without sacrificing precision.

11. Methodology Note

This analysis reproduces and extends the workflow from the uploaded notebook (Bank_Customer_Churn_Prediction.ipynb). The original notebook referenced a local file path; the standard public Churn_Modelling.csv dataset (10,000 rows, identical schema) was used to execute the pipeline end-to-end and generate the results in this report.